

KRITI GUPTA

M.Tech, IIT Delhi
Software Engineering • AI/ML
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PROFILE

M.Tech student at Indian Institute of Technology Delhi with a Computer Science background and experience spanning software development, machine learning, computer vision, and research. Author/presenter of three IEEE conference publications, with hands-on experience in computer vision-based systems and continued development in Data Structures & Algorithms, Machine Learning, and Software Engineering.

EDUCATION

Indian Institute of Technology Delhi	Jul 2026 – May 2028
M.Tech, Telecommunication Technology & Management (JTM)	
Vellore Institute of Technology, Chennai	2021 – 2025
B.Tech, Computer Science	
CBSE Class XII	96.2%
CBSE Class X	93.2% 100/100 Mathematics

ACADEMIC ACHIEVEMENTS

- **GATE 2026** — Score: **613** | AIR: **2763** | **98.69** percentile.
- School Mathematics topper; Class X Mathematics: **100/100**.
- CBSE Class XII: **96.2%**.
- CBSE Class X: **93.2%**.

RESEARCH & PUBLICATIONS

Review on Age and Gender Detection: Diverse Approach and Algorithmic Insights

- IEEE conference publication presented at the **2023 6th International Conference on Recent Trends in Advance Computing (ICRTAC)**.
- Conference: **December 2023, VIT Chennai**.
- Role: **Presenter and Researcher**.
- Research focused on approaches and algorithms for age and gender detection using computer vision and machine learning techniques.

HBTrack: AI-Based Heart Attack Prediction through ECG Monitoring on Wearable Devices

- IEEE conference publication presented at the **2024 First International Conference on Pioneering Developments in Computer Science & Digital Technologies (IC2SDT)**.
- Conference: **August 2024, NIT Delhi**.
- Role: **Presenter and Researcher**.
- Research explored AI-based heart attack prediction through ECG monitoring using wearable-device technology.

An Innovative Approach to Automate Nutritional Analysis Using Computer Vision and Machine Learning

- IEEE conference publication presented at the **2025 IEEE 4th International Conference on Technology, Engineering, Management for Societal impact using Marketing, Entrepreneurship and Talent (TEMSMET)**.
- Conference: **October 2025, NIT Delhi**.
- Role: **Presenter and Researcher**.
- Research focused on automating nutritional analysis using computer vision and machine learning.

PROJECTS

Nutritional Analysis using Computer Vision

Python, OpenCV, YOLO, Roboflow, Pandas

- Developed a computer vision system for detecting food and dairy products from images and webcam input.
- Implemented YOLO-based object detection through the Roboflow inference platform.

- Used OpenCV for image processing, webcam frame capture, and visualization of detection results.
- Integrated nutritional information lookup using a structured nutritional dataset and Pandas.
- Implemented bounding-box visualization for detected products and retrieval of corresponding nutritional information.
- Added nutritional value aggregation for detected products to support combined nutritional analysis.
- Implemented secure API-key configuration using environment variables rather than storing credentials directly in source code.

TECHNICAL SKILLS

- **Programming Languages:** Java (Fluent), Python (Working Knowledge), C (Working Knowledge)
- **Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming, Problem Solving, Git/GitHub
- **Machine Learning:** Regression, Classification, Logistic Regression, Gaussian Discriminant Analysis, Naive Bayes, SVM, KNN, PCA, Gaussian Mixture Models, EM, Model Evaluation, Regularization
- **Computer Vision:** Image Processing, Object Detection, CNNs, YOLO, OpenCV, Camera Calibration
- **AI/ML Interests:** Machine Learning, Computer Vision, Intelligent Systems

RELEVANT COURSEWORK

Computer Science / Software

Data Structures & Algorithms
Programming
Problem Solving

Communication / Systems

Digital Communication
Signal Processing Concepts

AI / Machine Learning

Machine Learning
Computer Vision
Artificial Intelligence / Intelligent Systems

Current Development

LeetCode Problem Solving
Data Structures & Algorithms Practice

EXPERIENCE & LEADERSHIP

Management Team — ResourceX, VIT Chennai

- Member of the management team of ResourceX at VIT Chennai, contributing to student-led organizational and event management activities.

Management Team — Women in Engineering, VIT Chennai

- Member of the management team of Women in Engineering at VIT Chennai.
- Contributed to student-led organizational and management activities.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner**
- **Google Cloud Digital Leader**
- **UI/UX Certification**

CURRENT DEVELOPMENT

- Actively practicing **Data Structures & Algorithms** and problem solving through LeetCode using Java.
- Building practical projects to strengthen **Software Engineering** skills.
- Developing knowledge in **Machine Learning and Computer Vision** alongside the M.Tech curriculum at IIT Delhi.
- Exploring the intersection of **Software Engineering, AI/ML, and Intelligent Systems**.

ADDITIONAL INFORMATION

- **Interests:** Software Engineering, Machine Learning, Computer Vision, Intelligent Systems
- **Languages / Communication:** English, Hindi